Theories Of Knowledge

Blithering Genius

2023 August 1

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1 Introduction

What is knowledge? How can an idea in a mind/brain represent something outside that mind/brain? How do images on a retina create the idea of a tree? When I see a tree, how do I recognize it as a "tree"? What is the abstract concept "tree"? How does this abstract concept relate to actual trees, and to my ideas of specific trees?

Those are all questions that a theory of knowledge should answer. In this essay, I will consider various theories of knowledge, and whether they can answer those questions.

In ordinary life, people take knowledge for granted. When they see a tree, they do not ponder how they know that the tree exists, or what the concept "tree" is. They just walk around the tree, or duck under its branches, and carry on with their lives. They leave those questions "to the philosophers" (and to the psychologists).

But if we want to understand ourselves, we cannot take those questions for granted.

2 What is Knowledge?

Knowledge is stored information that can be used to solve problems.

I define four types of knowledge:

- Procedural
- Conceptual
- Representational
- Formal

Procedural knowledge consists of innate mental processes, such as emotion, attention, induction, perception and cognition. Those processes "know" how to carry out mental operations. Procedural knowledge could be generalized to include all biological forms. We could say that a kidney "knows" how to filter the blood. But in this context, mental processes are the most relevant type of procedural knowledge.

Conceptual knowledge consists of concepts, such as "tree", and associated information. A concept contains (or is linked to) the information necessary to recognize instances of it, such as the visual pattern of a tree. Conceptual knowledge also includes relations between concepts, such as that trees have branches and wood comes from trees. Conceptual knowledge is subconscious. It depends on procedural knowledge for its acquisition and application.

Representational knowledge consists of ideas that are accessible to consciousness. In most cases, representational knowledge can be expressed in speech. Representational knowledge depends on conceptual knowledge and procedural knowledge. Representations (ideas/models) are constructed out of concepts by mental processes.

Most theories of knowledge focus on representational knowledge. Many do not even posit other types of knowledge, or conflate different types of knowledge. That is a major source of error.

It is natural that we focus on representational knowledge, because we are directly aware of ideas. We are not directly aware of concepts and mental processes. But to understand ideas, we must understand concepts and mental processes.

3 Examples Of Knowledge

In this essay, I will use two main examples of knowledge. One is the idea of the tree outside my window. The other is my knowledge of the English language. The first is representational. The second is conceptual.

If I look out my window, I see a tree. The mental process of perception automatically generates the idea of a tree in my mind. The idea is linked to my senses, but it is not purely sensory. My brain constructs it from the information of my senses, and from my background knowledge about the world. Mental processes recognize the tree (from sensory information) as an instance of the general concept "tree".

The application of a concept has implications. For example, I know that the tree outside my window has wood beneath the bark and roots below the ground, although I have no direct sensory information about those things. That knowledge comes from the concept "tree".

I learned the English language from the data of speech, and I use it to interpret and generate speech. My knowledge of language is conceptual. It consists of many concepts, including words and rules of grammar. Speech consists of ideas (words, sentences) that are generated by applying those abstract concepts to specific situations, just as my idea of the tree is generated by applying the concept "tree" to specific sensory data.

Language is a good example to illustrate the layers of knowledge. You have the innate ability to learn a language (procedural knowledge). You can know a language, such as English (conceptual knowledge). You can know a line from Hamlet, such as "To be, or not to be?" (representational knowledge).

The procedural knowledge of how to learn a language is innate and subconscious. The conceptual knowledge of a language is acquired and subconscious. The representational knowledge of a line from Hamlet is acquired and accessible to consciousness.

Language is also an example of cultural knowledge. Trees exist outside human brains, and do not depend on human brains for their existence. By contrast, language exists in human brains and has no independent existence. Language is part of culture. It exists as a concordance between different brains. My knowledge of English is a reflection of the same knowledge in other brains. Language is circular in that way.

4 What Is Formal Knowledge?

Now, let's talk about formal knowledge.

Formal knowledge is a special type of representational knowledge, which consists of explicit, *written* formulas, such as scientific laws and mathematical theorems. Formal knowledge is an attempt to make procedural or conceptual knowledge explicit. Like conceptual knowledge, formal knowledge is generic. It consists of general categories and rules, not referential ideas about specific things. It often has associated routines or algorithms, which resemble procedural knowledge.

"Trees contain wood" is a very simple example of formal knowledge. It is representational, but what it represents is a regularity of nature, not an object or event. It makes implicit conceptual knowledge explicit and representational.

Formal logic is a more complex example of formal knowledge. It represents something about how our brains work. An explicit grammar of the English language is another example. It represents (part of) the conceptual knowledge that is stored in the brains of English speakers.

Formal knowledge should not be confused with procedural or conceptual knowledge. Formal logic is not the same as the ability to think logically. The ability to think logically is innate, procedural knowledge. Formal logic is something that you can learn by reading a book or taking a course. Likewise, a formal grammar of English is not the conceptual knowledge that we use to interpret and generate speech.

A common error, in thinking about knowledge, is to assume that all knowledge is formal. People make this error because formal knowledge is explicit, so they are familiar with it. They assume that our knowledge of the world is stored in explicit statements, such as "Ravens are black". This is a serious error, which prevents intellectual progress.

Formal logic depends on procedural and conceptual knowledge. It presupposes logic and meaning. If we assume that procedural or conceptual knowledge is like formal knowledge, we are engaging in a type of question-begging — a subtle homunculus fallacy. "Ravens are black" has no meaning without procedural and conceptual knowledge. We can't understand formal knowledge until we understand the deeper layers of knowledge.

Again, it is important to make the distinction between different types of knowledge, and understand how they are related.

This has been a brief sketch of my theory of knowledge, with a few things left out. At the end of this essay, I will fill in the missing details. This sketch was necessary to understand the topic, and to understand my criticisms of other theories.

Now, let's consider some other theories of knowledge. I will group them into broad categories, and define them in my own terms.

5 Naive Realism

As I define it, naive realism is the belief that truth is objective, and consists of a *correspondence* between belief and reality. A belief is true if it corresponds to reality, and false otherwise. Knowledge consists of true beliefs.

Note that "realism" does not have its ordinary meaning here. There is nothing realistic about naive realism.

Naive realism is naive because it simply ignores the philosophical problem of knowledge. How does an idea in the mind/brain represent something else? A theory of knowledge should answer that question, but naive realism ignores or dismisses it. It just assumes that ideas in the mind somehow correspond to facts in objective reality.

Naive realism is the ordinary person's implicit theory of knowledge. People talk about "the facts" and what is "objectively true", without thinking about the meanings of those terms. They presuppose that ideas can correspond to reality – somehow – and that we can establish truth or falsity by comparing a belief to "the facts".

But what are facts? In what sense do they exist? How can we compare ideas in our brains to objective facts?

The naive realist has no answers to those questions, because he hasn't thought about the nature of knowledge or the relation between ideas and reality.

Naive realism is not a satisfactory theory of knowledge. It's just a collection of hidden assumptions that have never been questioned. It takes knowledge for granted.

Some philosophers have attempted to make naive realism more explicit and less naive, but they have failed. Such attempts just beg the question explicitly rather than implicitly.

For example, Bertrand Russell defined truth as follows:

> We may restate our theory as follows: If we take such a belief as “Othello believes that Desdemona loves Cassio”, we will call Desdemona and Cassio the object-terms, and loving the object-relation. If there is a complex unity “Desdemona’s love for Cassio”, consisting of the object-terms related by the object-relation in the same order as they have in the belief, then this complex unity is called the fact corresponding to the belief. Thus a belief is true when there is a corresponding fact, and is false when there is no corresponding fact.
>
> – Bertrand Russell “On Truth and Falsehood”

But what is a fact? It is just a true belief. Russell assumes that something exists in reality (“a complex unity”) which has the same structure as a belief. That is what he calls “the fact corresponding to the belief”. Nothing is gained by this silly definition. It is circular. It ignores the question of how an idea in the mind relates to reality.

Naive realism ignores the difference *in kind* between ideas in the mind/brain and what they represent. Objective facts would (presumably) have the same form as ideas, and thus ideas could correspond to facts. But why would there exist something outside a brain that has the same form as an idea? And even if such facts exist, how would ideas in the mind correspond to them? What is the mechanism of correspondence?

Naive realism does not answer those questions.

6 Subjective Idealism

As I define it, subjective idealism is the belief that objective reality either does not exist or is unknowable. Subjective idealism claims that we should not posit objective reality, since we have no access to it. We only have access to the content of our minds. Whereas naive realism denies or ignores the difference in kind between ideas and reality, subjective idealism rejects reality altogether.

Note that “idealism” does not have its ordinary meaning here. Subjective idealism could also be called “solipsism”.

Subjective idealism comes from radical skepticism about truth and knowledge. The skeptical arguments against knowledge (I could be dreaming, I could be hallucinating, I could be a brain in a vat, etc.) show that beliefs about reality are never beyond doubt. I can doubt anything about the external world, including the existence of an external world. When I perceive a chair, I can doubt the existence of an object that I am perceiving. However, I cannot doubt the existence of the idea of the chair, because I directly experience it. Knowledge claims about objective reality are dubious and unjustified, while claims about subjective experience are indubitable and intrinsically justified.

There is a subtle error involved here, however. The perception of a chair *presupposes* objective reality. The idea of a chair (generated by perception) is *about* objective reality. I subjectively experience the idea as a representation of objective reality.

Subjective idealism does not answer the question of how ideas relate to reality. Instead, it rejects a presupposition of the question: the existence of objective reality.

Subjective idealism is untenable, because (a) We naturally presuppose objective reality, and (b) We experience phenomena that are best explained by positing objective reality.

Positing objective reality helps to explain the following:

- The vividness of perception, and the difference between perception and memory/imagination. The vividness of perception is explained by the novel information of sensory data, coming from external reality.
- Surprise and confusion. If there is no external source of information, how could I be surprised or confused?
- Learning from experience, such as learning a language. If there is no objective reality, where does the information come from?
- Errors of judgment, which are corrected by more information or thought. If there is no objective reality, how can an idea be wrong?
- The difference between walking into a strange room and walking into a familiar room. If there is no external reality, where does the new information come from?

These phenomena are best explained by positing an objective reality that is distinct from our ideas of it. Our knowledge of reality is limited and imperfect, because it is constructed by mental processes from sensory data.

The concept of knowledge presupposes the subject | object distinction. Thus, no theory of knowledge can reject that distinction.

Subjective idealism is not a satisfactory theory of knowledge. It does not explain how ideas in the mind/brain relate to reality.

7 Objective Idealism

As I define it, objective idealism is the belief that objective reality consists (partially or wholly) of ideas. It does not reject the subject | object distinction, but it rejects the distinction *in kind* between ideas and objects. It claims that there are objective ideas, and that truth is a correspondence between a subjective idea and an objective idea.

Note that "idealism" does not have its ordinary meaning here.

In academic philosophy, objective idealism is called "metaphysical idealism" or "ontological idealism". It is also related to panpsychism (All is mind) and pantheism (All is God). If reality consists of ideas, then presumably there is a universal mind in which those ideas exist (the mind of God). If reality consists of ideas, then perhaps consciousness exists in all things, not just in special objects (brains of a certain type).

Objective idealism is an attempt to rescue the correspondence theory of truth: that a true belief corresponds to an objective fact. It is hard to understand how the idea of a tree, consisting of some mental state in a brain, could correspond to an actual tree. But if the tree is also an idea, then perhaps the two could correspond.

Objective idealism does nothing to solve the philosophical problems of knowledge. It just shuffles them around. It replaces the question "How do ideas in the mind relate to objective reality?" with "How do ideas in the mind relate to objective ideas?". Nothing is explained by positing that objective reality is a mind or consists of mind-stuff.

If objective ideas existed, they would be different from subjective ideas. Otherwise, we couldn't explain the imperfect and limited nature of knowledge.

For example, my idea of the tree outside my window does not include the knowledge of how many branches it has, or how deep its roots go, or its mass to the nearest milligram. Presumably, the objective idea would contain all of that information and much more. Thus, the two ideas would not be identical. How could we say that one corresponds to the other? At most, one would be a very rough and distorted approximation of the other.

In physics, we represent a planet as a point mass, and the motions of 10^{24} molecules as a temperature. You can use a street map to represent a city. We do not believe that these models are identical to what they represent, even in the information that they contain. They are much simpler.

Even if objective ideas existed, our subjective ideas would not correspond to them. Nothing is explained by positing objective ideas, and the correspondence theory is not rescued.

Most objective idealists believe that objective reality is a mind — the mind of God — and that objects and events are ideas in God's mind. This is an inverse homunculus metaphor. (See "The Homunculus Fallacy and its Inverse".)

Some objective idealists claim that their theory explains human consciousness. But this is false. Even if we posit a cosmic mind, that would not explain anything about human consciousness. We would still have to explain how the human mind depends on the brain: how we fall asleep and wake up, how a pill can make you unconscious, how brain activity correlates with feelings and ideas, how brain injuries affect memory and personality, etc. We would still have to explain how ideas in the human mind relate to objective reality (ideas in the cosmic mind). We would still have to explain all the phenomena that subjective idealism cannot explain, such as learning, surprise, the vividness of perception, etc.

Calling objective reality "a mind" explains nothing. It is just a confusing metaphor.

Also, the claim "all is mind" is vacuous. If everything is mind, then "mind" is meaningless. On the other hand, if there is a distinction in kind between the human mind and reality, then why analogize reality to the human mind? Why not just call it "objective reality", instead of "the cosmic mind"?

Nothing is explained by objective idealism.

8 Transcendentalism

As I define it, transcendentalism posits another realm that transcends both objectivity and subjectivity. This realm contains ideas, concepts or metaphysical presuppositions. Somehow, it links subject and object.

Plato was an early proponent of transcendentalism. He noticed that a statement such as "There are three horses in that field" invokes the abstraction "horse", which is actualized by three objects. He believed that the abstract form of a horse was non-physical (metaphysical?), and that it transcended space and time. He believed that this abstract form was somehow related to the objects that actualize it. He also believed that our knowledge of reality involves the recognition of these abstract forms.

Our word "idea" comes from the Greek "", which means "form". Apparently, both "view" and "idea" derive from the Indo-European root "weid". See Online Etymology Dictionary: idea.

However, Platonic forms correspond to concepts, not ideas (as I use the terms). When you see a horse, your brain uses the abstract concept of a horse to generate the idea of a specific horse.

Plato never explained how the transcendental forms relate to objectivity or subjectivity. So, his theory has no explanatory power. Positing another realm doesn't explain anything. We should give Plato credit for making the distinction between abstract forms and specific instances of those forms, but he didn't explain how they are related, or what the abstract forms are.

Plato's theory seems very silly today, in light of modern science. We know how the recurring pattern "horse" emerged in the world (by evolution), and that it isn't eternal or unchanging. Some forms, such as the laws of physics, might transcend space and time. We could say that they exist "on a higher plane" or in a "different realm of existence", metaphorically. But most of our concepts reflect order that is emergent and transient, not ultimate and eternal.

Plato's theory led to a long debate over whether "universals" (forms/concepts) are just names (nominalism) or are objects in some other realm (Platonism). Apparently, the participants in this debate never considered other possibilities.

These days, Platonism is mostly confined to the philosophy of mathematics. Some believe that mathematical theorems exist in another realm (the realm of mathematics), and that we have mental access to this realm, somehow. When we discover a mathematical theorem, we are discovering something that exists in this other realm.

This has no explanatory power. It also ignores that we create somewhat arbitrary things, such as language, that don't exist as physical objects, but consist of *information*. The works of Plato, including his theory of forms, belong to that category. Mathematics is less arbitrary than the English language, and more general than Plato's philosophy, but it is still a human creation.

This brings us back to the different types of knowledge: procedural, conceptual, representational and formal. We construct ideas. We induce concepts. We are born with innate mental faculties. Our mental faculties have *presuppositions* built into them. We can develop explicit theories of those presuppositions. Such theories are formal knowledge.

Our mental processes presuppose space, time, causality, identity, logic, number and the subject | object distinction. Space is automatically involved in perception, thought and action. You can develop the concept of space secondarily, but that is not the same as the innate presuppositions about space that are built into your brain. Those presuppositions are implicit in the structure of your brain, and are manifest in perception, thought and action.

We could say that the presuppositions of subjectivity "transcend" subjectivity, because they are its necessary antecedents. We can't think without them. However, it does not follow that they exist in a different realm, or that they exist independently of us.

We model objective reality within the framework defined by those presuppositions. There is no way for us to know what objective reality ultimately consists of. All we can do is model it in the framework provided by our brains.

There are different types of transcendentalism, which place different types of knowledge into the transcendental realm. This often involves a confusion between different types of knowledge, such as confusing formal knowledge with procedural knowledge (e.g. confusing formal logic with innate logic).

Some Christians believe that God's mind is a transcendental realm that links subject and object.

In this theory, mental presuppositions are derived (somehow) from God's mind. We acquire knowledge of the material world from experience, but we have access to logic, mathematics and other procedural/presuppositional knowledge via our connection to God. Typically, they think of this transcendental knowledge as having a propositional form, like formal knowledge, rather than being structurally implicit, as procedural knowledge is.

Transcendentalism does not answer any philosophical question about knowledge. It posits another realm of existence, which supposedly bridges the gap between subject and object, but it does not explain how this realm relates to subjectivity or objectivity.

9 Consequentialism

I will use the term "consequentialism" for theories that define truth and knowledge as "whatever works". In academic philosophy, such theories are labeled "pragmatism". However, there is nothing pragmatic about them, and the term "consequentialism" is more descriptively accurate.

In consequentialism, knowledge is not viewed as a correspondence to reality, but as a means to effective action. Truth (the correctness or goodness of knowledge) is defined in terms of utility. If an idea is useful, then it is "true" for practical purposes.

Consequentialism is not a satisfactory theory of knowledge. It does not explain how ideas in the mind relate to objective reality. It also begs the question of knowledge.

Suppose that a brain makes judgments of truth using the criterion of consequentialism. It selects ideas that would have beneficial effects if believed. How could it do that? How could the brain *know* whether an idea would have beneficial effects? The utility of an idea depends on how it interacts causally with both the brain and objective reality. To judge the utility of an idea, the brain would need to make truth judgments about its effects. "Belief X would have utility if believed" is itself a belief.

Let C(X) denote the proposition that belief X would have utility to the believer, and thus is considered "true" by the consequentialist criterion. Suppose that I am a consequentialist, trying to make a truth judgment about a proposition P. To decide whether P is true, I must decide whether C(P) is true. To do that, I must decide whether C(C(P)) is true. To do that, I must decide whether C(C(C(P))) is true. And so on.

Consequentialism is an infinite regress. Truth judgments would depend on truth judgments to infinity. Somewhat ironically, this "pragmatic" method of making truth judgments is utterly useless.

Truth judgments cannot be based on value judgments, because value judgments depend on truth judgments. We need a method of making truth judgments before we can make value judgments.

Knowledge is pragmatic in the sense that it has a biological function. Our brains evolved to use knowledge to generate adaptive behavior. Induction evolved to generate useful knowledge. However, induction cannot select knowledge based on its predicted utility.

From an external perspective, we can judge a person's beliefs by the consequentialist criterion. For example, I could look at an Amish farmer and say:

> Well, his religion is absurd and false, but it is part of a functional worldview. He is reproducing. If his beliefs have reproductive utility to him, they are correct in a sense.

However, I can make that judgment only because I am capable of making truth judgments by another method. To make that judgment about his religious beliefs, I need to predict their effect on his reproduction. That prediction is not based on the consequentialist criterion. Note that I can judge his beliefs to be false while affirming their utility to him.

Our brains have criteria built into them, by which they induce conceptual knowledge and generate ideas. Those criteria work, and they are normative in a sense. But they are not value judgments, and we do not derive truth judgments from value judgments.

Consequentialism does not describe the relation between ideas in the mind and reality, or how ideas are selected by the brain. Its definition of truth is circular.

10 Intersubjective Idealism

As I define it, intersubjective idealism is the belief that truth and knowledge are intersubjective. To the intersubjective idealist, truth is a correspondence between ideas in different minds, rather than a correspondence between ideas and objective reality. This implies that truth is socially relative. Intersubjective idealists will say that truth is a "social construct".

Intersubjective idealism is not a satisfactory theory of knowledge, because it does not explain how ideas in the mind relate to objective reality. If knowledge is just a relation between minds, then knowledge is vacuous. What is the point of minds having ideas, if those ideas have nothing to do with reality? Intersubjective idealism would render ideas meaningless.

Intersubjective idealism does not normally stand alone as a philosophical theory of knowledge. However, it is an important part of post-modern leftism.

For example, leftists will claim that race and sex are social constructs. It is not entirely clear what they mean by this. Do they mean that the concepts of race and sex are socially influenced, and thus suspect? Or do they mean that race and sex are purely social categories, with no relation to objective reality? If those concepts have no relation to objective reality, how do we recognize instances of them?

A charitable interpretation is that there is a biological basis for the categories, but their discrete nature is imposed on a more complex reality. However, leftists say things like "trans-women are real women" and "gender-affirming healthcare", which presuppose that sex is entirely social, and that individuals have the social right to construct their own "gender".

Current gender ideology presupposes both that gender is biological and that it is an arbitrary social construct. See The Trans Paradox.

Leftists often seem to view truth and knowledge as merely an expression of social power.

Some things are socially constructed. Rights and obligations are social constructs. To define rights and obligations, we use social constructs such as property, money, laws, citizenship, borders, marriage, promises, etc. We construct all of those things by performing certain conventional behaviors. For example, you can construct a promise by saying "I promise to do X". The statement creates a promise, which implies a right and an obligation.

Together, social constructs constitute *social reality*. Rights and obligations are real, but not objective. They are intersubjective. Social reality is socially constructed.

Social construction is an important source of certain ideas. However, it is not the source of all ideas. And social constructs derive their meaning from their relations to other ideas, and ultimately from their relations to objective reality.

For example, money is a social construct, but you can use it to buy a hamburger. The idea of buying a hamburger involves the social construct of money, but it also involves ideas about objective reality, such as the idea of a hamburger. If your friend promises to help you move to a new apartment, the promise is intersubjective, but the physical event of moving is not. We can't make hamburgers or move boxes just by having shared ideas. Social reality "cashes out" in objective reality.

There is also social knowledge, which is different from social reality. Like individuals, societies need to acquire knowledge and make truth judgments. We have social methods of knowledge acquisition and truth judgment, such as science. Those methods combine the knowledge and judgment of many individuals. Social knowledge and social truth are social, but not *entirely* social. They depend on individual mental processes, and they depend on objective reality.

A scientific theory and a tax code are both socially created, but not in the same way. The scientific theory is socially learned. The tax code is socially constructed.

The scientific theory depends on objective reality through observation, experimentation and application. It explains and predicts certain aspects of objective reality. It reflects regularities of nature. We could use it to design technology, make individual choices, or select social policies. But the theory itself is descriptive, not normative.

The tax code emerges out of the interplay of individual desires. It does not represent anything in objective reality. It is merely an agreement between minds, which reflects a balance of power. The tax code is a way to organize human behavior. It is socially normative.

We have social processes for creating scientific theories and tax codes, and they have different criteria. A scientific theory is created by the scientific method. A tax code is created by a political process. Those processes are both social, but not in the same way.

We don't exist as isolated individuals. We are connected to other minds through communication and social interaction. We acquire knowledge from culture. Knowledge and truth claims have social functions, and can be influenced by social pressures. Society creates the potential for delusion, deception and fiction.

Society and culture exist, and are important in human affairs. It does not follow that we can discard the concepts of objective reality and the individual. Society is composed of individuals. Social knowledge and social reality depend on individual brains. Social knowledge depends on objective reality. Social knowledge and social reality both "cash out" in objective reality.

11 Representationalism

Now, I will flesh out my theory of knowledge, under the label "representationalism". Apparently, this label is used for theories like mine, but I will only present my own theory. I might not agree with other theories that have the same label.

The relation between ideas and reality is *representation*, not correspondence. An idea in the mind can represent an object in reality, such as the coffee cup on my desk. Ideas represent objective reality in a subject-dependent way: in terms of how the object relates to the subject.

The brain induces concepts, such as “coffee cup”, from what I call “semex”. Semex consists of sensory, emotional and motor data. Concepts are abstractions from semex, and they reflect regularities in semex. Induction is based on the criterion of *information compression*, which is implicit in mental processes.

Once concepts have been induced, they can be used to interpret experience and generate action. The process of perception uses the concept “coffee cup” to model my current situation. My idea of the coffee cup on my desk is an application of the abstract concept of a coffee cup. That abstract concept contains information about how I can interact with coffee cups: how they affect my senses and emotions, and how I can affect them.

The idea of the cup fits the data of my current semex. It explains part of my visual experience. It implies the potential action of drinking coffee from the cup. It predicts the sensory and emotional consequences of that action.

The idea represents the coffee cup to me. It links subject and object, and it depends on both.

In this theory, there is order to reality, and there is order to my brain. The brain has the procedural knowledge of how to induce concepts from experience. Those concepts depend on the order of objective reality, because semex depends on objective reality. But the concepts are not the order of objective reality itself (the form-in-itself?). A concept is a regularity of human interaction with the world.

The presuppositions of my mental processes are implicit in the form of my brain, which evolved by natural selection. I have innate procedural knowledge, but my conceptual and representational knowledge depends on experience.

In this theory, truth is not a correspondence between an idea and reality. It is a judgment made by a brain. If a model is selected to represent reality by my brain, then that model is true to me. To call a claim “false” means that it does not agree with my truth judgment. Truth is a subjective judgment.

Information compression is the underlying norm of truth. It is implicit in my mental processes. The brain selects concepts that compress the data of past experience. It applies those concepts to semex based on how well they fit the data of semex. Fit to data is defined as the degree to which a model reduces the information of the data.

Knowledge is pragmatic in the sense that it is biologically functional. But this norm does not reduce truth to value. Information compression is defined relative to the data of semex. Our brains do not induce knowledge or make truth judgments based on predicting their consequences. That would be an infinite regress. Information compression is not an infinite regress.

The subjectivity of truth doesn’t make it random or whimsical. Truth judgments depend on brains, and brains are ordered mechanisms. The form of the brain was selected by evolution to generate adaptive behavior, and every component of that form has a function relative to the function of the whole.

We can develop a theory of those forms and functions, and that theory is formal knowledge. Based on that theory, we expect truth to be roughly convergent for different people, and we can define norms (such as principles of rationality) and common error patterns (fallacies) that apply to human beings generally, because they reflect the functions and malfunctions of mental processes.

Ultimately, we are linked to the external world and other people through *order*. The brain is ordered, and the world is ordered.

What is order? The compressibility of information. Regularity. The similarity of one thing to another. The similarity of one moment of experience to another.

The form of a horse does not exist in some external realm. However, it does exist as a recurring pattern in nature, due to evolution. Evolution is a deeper regularity, and it arises from even deeper regularities. Just as we have no direct access to the thing-in-itself, we have no direct access to the order of objective reality. But we can represent it with abstractions, such as the concept “horse” or the theory of evolution.

Concepts in one brain are often influenced by concepts in other brains. Some conceptual knowledge is purely cultural. Language is a good example. The English language isn’t knowledge of something else. It is a culturally emergent system of concepts that we use to communicate ideas.

Some ideas are socially constructed. The construction of social reality is consequentialist. People try to construct social reality according to their desires. But that is a special case, and social reality depends on knowledge of objective reality.

Because we live in the same world and have similar brains, we develop (roughly) the same concepts from experience. This allows us to mentally construct similar ideas. Language (based on shared procedural and conceptual knowledge) allows us to communicate ideas, and it causes our conceptual knowledge to be even more convergent. We relate to each other through shared ideas.

Consider buying a cup of coffee, for example. When I walk into a coffee shop, the barista and I have a shared model of the situation. I don’t need to tell her that I want to buy something, or explain what “buying” is, or negotiate an acceptable medium of exchange. That background knowledge is present in the shared model. I just tell her what items I want, and provide payment. Together, we socially construct the purchase of a cup of coffee in social reality, and I get my coffee in objective reality.

There is no need to posit a magical realm, or God’s mind, to mediate between individuals or between the individual and reality. We are connected to reality through our senses and muscles. We are connected to each other through culture, biology and the world that we live in.

Representationalism is based on important distinctions:

- Procedural, conceptual, representational and formal knowledge.
- Processes, concepts and ideas.
- Subject, object and idea.
- Truth, value and action.
- The individual and society.
- Objectivity, subjectivity and intersubjectivity.

Representationalism explains:

- How the brain acquires knowledge of reality from the data of embodied experience.
- How a concept, such as “coffee cup” or “tree”, relates to objective reality.
- How an idea in the mind, such as “the tree outside my window”, relates to an object in reality.
- How the brain uses knowledge to make decisions, both consciously and subconsciously.

- How we use language to communicate.
- How we use social constructs to organize social interaction.
- How we generate social knowledge and make social judgments of truth.